Label-Efficient Dark-Vessel Detection in SAR:

Does SAR-Domain Pretraining Beat Optical — and Does It Hold Across ViT and CNN?

John Roth & Kyle Wagner · EN.705.643 Deep Learning Developments with PyTorch

1 Problem and Motivation

Detecting ships that are not broadcasting their automated identification system (AIS) signature — "dark vessels" — in spaceborne synthetic aperture radar (SAR) is a critical challenge in maritime monitoring, largely because labeled data is scarce. The xView3-SAR benchmark provides Sentinel-1 imagery for this task, but its verified dark-vessel labels are few, so relying solely on supervised training often leads to overfitting. Pretrained foundation backbones can help, but which type transfers best — generic optical, SAR-specific, or labeled out-of-domain SAR — is unclear, and whether the answer depends on the backbone's architecture is unknown. We evaluate both questions by comparing downloaded pretrained backbones across two architecture families under a fixed detector, making initialization and architecture the only variables. Using existing checkpoints rather than pretraining our own keeps compute low and the focus on a metric-matched evaluation.

2 Research Question

Under one fixed detection harness, how much labeled xView3 data does each class of pretrained backbone save for dark-vessel detection; does a SAR-domain foundation model beat an optical one in the scarce-label regime; and does that finding hold across two architecture families (ViT and CNN)? **Hypothesis:** within each architecture, SAR-domain pretraining yields the steepest label-efficiency gains and the largest dark-vessel-recall advantage at low budgets (ordering, per track, roughly random $<$ optical $\lesssim$ supervised $<$ SAR, converging as labels grow), and this ordering is *architecture-general* — the SAR-vs-optical gap appears for both the ViT and the CNN. The magnitude of that gap quantifies the value of domain-matched pretraining.

3 Approach

The detector pairs a backbone with a heatmap detection head: each vessel is a Gaussian blob on a stride-4 response map, decoded by peak-finding with distance-based non-maximum suppression. This is point-native: xView3's labels are points and its official metric matches predictions to ground truth by distance (not box IoU), so a heatmap-and-distance formulation mirrors the metric exactly and avoids synthesizing bounding boxes the data never had. The study has **two architecture-matched tracks** — a ViT track (ViT-B/16, ∼86M) and a CNN track (ConvNeXt-V2-Base, ∼89M, deliberately size-matched). **Fairness by construction:** within each track all four arms share the same backbone, head, fine-tuning schedule, seeds, and the fixed input ([VH, VV, VH−VV] in dB); only the downloaded initialization differs. The head, optimizer, and schedule are shared across both tracks, so ViT-vs-CNN is a fair comparison. Every pretraining claim is made *within* a track (architecture fixed); every architecture-generality claim compares *matched roles across* tracks (pretraining role fixed). We report parameter count and GPU-hours per arm. Data is xView3 via the SARFish mirror (Sentinel-1 GRD, scene-level splits); LS-SSDD-v1.0 is the labeled source for the two supervised arms.

Arm	Backbone	Initialization	Role
Controlled study — two architecture-matched tracks (plotted on the label-efficiency curves)			
1	ViT-B/16	random init	ViT floor
2	ViT-B/16	SatDINO (fMoW-RGB, DINO)	optical-domain FM transfer
3	ViT-B/16	SARMAE (SAR-1M, MAE)	SAR-domain FM transfer
4	ViT-B/16	LS-SSDD supervised (we train)	labeled SAR-detection transfer
5	ConvNeXt-V2-B	random init	CNN floor
6	ConvNeXt-V2-B	BigEarthNet-S2 (optical RS)	optical-RS transfer
7	ConvNeXt-V2-B	BigEarthNet-S1 (SAR RS)	SAR-RS transfer
8	ConvNeXt-V2-B	LS-SSDD supervised (we train)	labeled SAR-detection transfer
Reported separately (not on the curves)			
R1	ConvNeXt-V2-B	ImageNet (contingent, if time)	generic-transfer reference
R2/R3	YOLO26 / LocateAnything	COCO / zero-shot	external reference range

Tracks are matched by role (floor 1/5, optical 2/6, SAR 3/7, supervised 4/8); comparing arm k to k+4 isolates architecture with pretraining fixed.

The downloaded backbones are, for the ViT track, SatDINO (optical; DINO self-distillation on fMoW-RGB; Straka & Gruber 2025) and SARMAE (SAR; masked autoencoder on the million-image SAR-1M set; Liu et al., CVPR 2026); and for the CNN track, ConvNeXt-V2-Base checkpoints from BigEarthNet v2.0 (reBEN; Clasen et al. 2025) in Sentinel-2 (optical) and Sentinel-1 (SAR) variants. We pretrain none of them; the only training we run is a cheap supervised-detection backbone per track on LS-SSDD (Arms 4 and 8). We state the cross-family caveats openly. The two ViT arms differ in self-supervised method (DINO vs. MAE) as well as domain — no downloadable optical ViT-B MAE existed to match SARMAE's method, so we compare the strongest released model of each domain as-is. The two CNN RS arms are instead the *cleanest* domain contrast in the study: identical model, dataset, and supervised task, differing only in input modality (SAR vs. optical); the CNN RS checkpoints are supervised-classification pretrained rather than self-supervised, a documented track-level property, not a within-track confound. On input: every backbone was pretrained on a 3-channel-ish input that is not dual-pol SAR; Sentinel-1 supplies two native polarizations (VH, VV), and we form a 3-channel tensor by adding their difference, [VH, VV, VH−VV], so each backbone's stem adapts to the same fixed input during fine-tuning via the standard `timm` channel-adaptation path — a bounded gap applied identically to every arm. This isolates optical-vs-SAR pretraining *and* its architecture-generality on dark-vessel, scarce-label detection, which to our knowledge has not been done. All training uses one PyTorch Lightning code path, so seeding, mixed precision, and distributed training are identical across arms.

4 Evaluation Plan

We split xView3 training scenes 75/15/10 at the *scene* level (no chip-level leakage); the ∼50 human-verified validation scenes are reserved and touched once, at final evaluation. The headline deliverable is a **two-track label-efficiency curve**: detection F1 versus 10/25/50/100% of real labels, eight lines (solid ViT, dashed CNN; one color per pretraining role; Figure 1). Within each track the 10%-label gap over the floor quantifies pretraining value and the SAR-vs-optical ordering; across tracks, matched-role gaps test whether the SAR advantage is architecture-general. We also report *dark-vessel recall* and *close-to-shore F1* — the hardest slices, where any SAR advantage should concentrate.

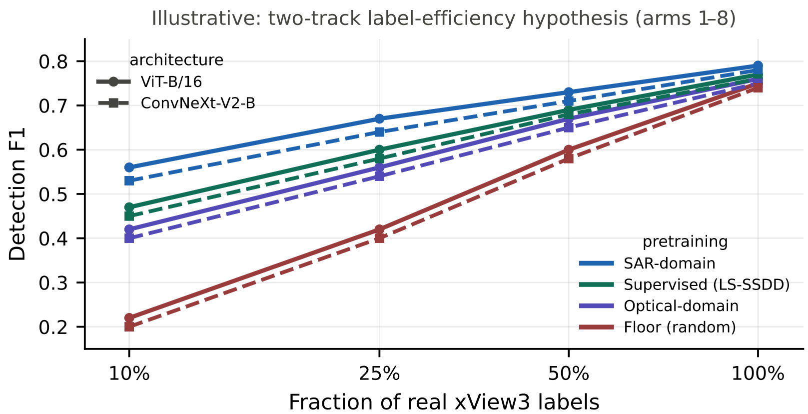

Figure 1: Illustrative two-track label-efficiency hypothesis (eight arms). Solid = ViT, dashed = ConvNeXt-V2; color = pretraining role. The SAR-vs-optical ordering is hypothesized to hold for both architectures.

5 Timeline, Compute, and Feasibility

The project runs from July 1 to the September 1 deadline, about nine weeks. Two contributors work largely independent lanes (data/scorer; harness/evaluation) with no hard dependency, since the foundation models are downloaded and there is no pretraining handoff to block on. Compute is one 8×V100 (32 GB) node plus an RTX 5070 Ti for development. Because we pretrain no foundation model, the total stays modest: each fine-tune is a few GPU-hours and the full eight-arm grid (both tracks, with seed reruns) plus the two small LS-SSDD supervised backbones comes to roughly **245–310 GPU-hours**, about two to three nights on the node. The downloaded-backbone arms run *first*, since the main technical risk is the channel-format gap (whether each pretrained stem accepts the [VH, VV, VH−VV] input); a quick load-and-train check de-risks the shared input per architecture before the full grid, and we confirm every dataset and checkpoint opens before committing. Given the focus on SAR foundation models and label-efficient detection, we will target a remote-sensing workshop such as CVPR EarthVision or ICCV.

Dates (2026)	Milestone
Jul 1–14	SARFish/LS-SSDD download, chipping, scene-level splits; sacred scorer + harness
Jul 15–28	Both tracks' FM + floor arms (ViT: SatDINO, SARMAE; CNN: BigEarthNet S1/S2); channel-format check; references
Jul 29–Aug 11	Supervised-transfer arms (ViT+CNN on LS-SSDD); full label-fraction grid; seed reruns
Aug 12–25	Final verified-scene eval; ViT-vs-CNN and dark-vessel/shoreline analysis; begin writeup
Aug 26–Sep 1	Finish writeup and submit (contingent ImageNet-ConvNeXt reference only if time remains)

Key References

[1] Paolo et al. *xView3-SAR*, NeurIPS 2022. [2] Straka & Gruber. *SatDINO*, 2025 (arXiv:2508.21402). [3] Liu et al. *SARMAE*, CVPR 2026 (arXiv:2512.16635). [4] Woo et al. *ConvNeXt V2*, CVPR 2023. [5] Clasen et al. *reBEN: Refined BigEarthNet*, IGARSS 2025 (arXiv:2407.03653). [6] Zhang et al. *LS-SSDD-v1.0*, 2020. [7] Zoph et al. *Rethinking Pre-training and Self-training*, NeurIPS 2020. [8] Jocher et al. *Ultralytics YOLO26*, 2026.